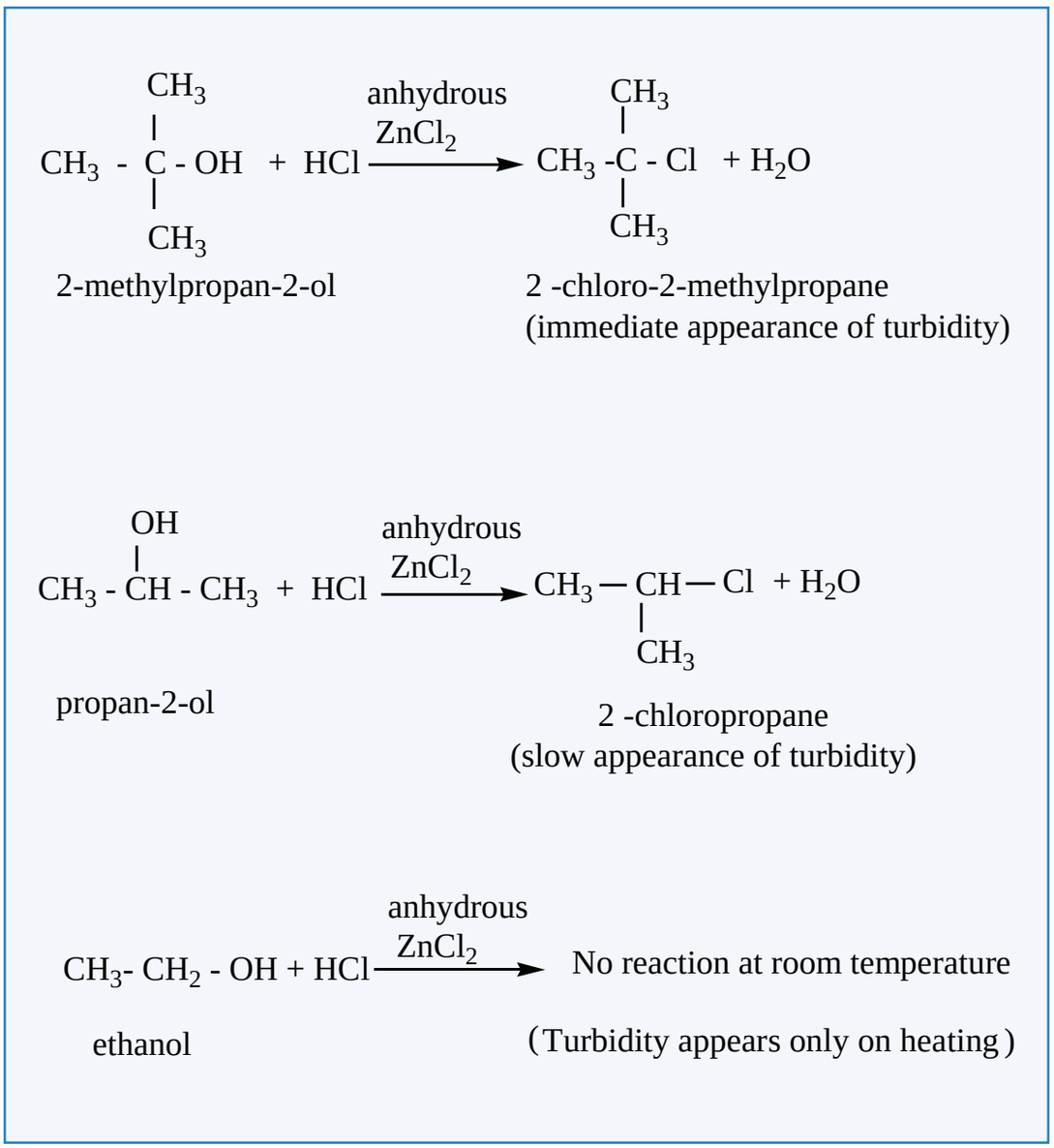

2-methylpropan-2-ol

2 -chloro-2-methylpropane
(immediate appearance of turbidity)

propan-2-ol

2 -chloropropane
(slow appearance of turbidity)

ethanol

No reaction at room temperature

(Turbidity appears only on heating)

b) Victor Meyer's test:

This test is based on the behaviour of the different nitro alkanes formed by the three types of alcohols with nitrous acid and it consists of the following steps.

i) Alcohols are converted into alkyl iodide by treating it with I_2/P.

ii) Alkyl iodide so formed is then treated with $AgNO_2$ to form nitro alkanes.

ii) Nitro alkanes are finally treated with HNO_2 (mixture of $NaNO_2$ / HCl) and the resultant solution is made alkaline with KOH.

Result:

- Primary alcohol gives red colour • Secondary alcohol gives blue colour.
- No colouration will be observed in case of tertiary alcohol.

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